

Multimodal NIFTY-50 Market Prediction System using Deep Learning

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Abstract—Accurate prediction of stock market indices remains a formidable challenge owing to the inherently noisy, non-stationary, and nonlinear nature of financial time series. This paper presents a multimodal deep learning framework for predicting the NIFTY-50 index of the National Stock Exchange of India. The proposed system integrates three heterogeneous data modalities—historical price sequences, fifteen computed technical indicators, and natural language sentiment derived from financial news and Reddit discussions—through a novel *Adaptive Fusion Gate* with learnable temperature-scaled softmax weighting. A Temporal Convolutional Network (TCN) with six levels of dilated causal convolutions encodes the price stream across a receptive field of 253 trading days; a batch-normalised MLP encodes the technical indicators; and a lightweight feed-forward network encodes the sentiment signals. The Adaptive Fusion Gate dynamically re-weights each modality according to prevailing market conditions, producing a unified 384-dimensional representation that is decoded by a multi-output head generating both a point forecast and quantile estimates at the 5th, 50th, and 95th percentiles, supplying calibrated uncertainty bounds alongside every prediction.

Beyond predictive accuracy, the system embeds SHAP (SHapley Additive exPlanations) directly into the inference pipeline. For each forecast, SHAP decomposes the model output into signed, feature-level contributions spanning all three modalities, making explicit which price patterns, technical signals, or sentiment shifts drove the decision. This transforms an otherwise opaque deep network into a transparent decision-support tool that meets the interpretability expectations of practising traders and regulated financial institutions. The complete system is deployed as a production-ready web application—FastAPI backend, Next.js 14 dashboard, PostgreSQL persistence—bridging the gap between academic research and real-world financial decision support.

Experimental evaluation on three years of NIFTY-50 trading data demonstrates an RMSE of 0.847%, MAE of 0.623%, MAPE of 2.41%, and directional accuracy of 62.8%, outperforming unimodal baselines by 8–15% and static fusion by 3.6%. Empirical coverage analysis confirms that 88.3% of actual returns fall within the predicted 90% interval, indicating well-calibrated uncertainty estimates.

Index Terms—stock market prediction, deep learning, multimodal fusion, temporal convolutional network, sentiment analysis, NIFTY-50, explainable AI, technical indicators, adaptive fusion gate, quantile regression, financial forecasting

I. INTRODUCTION

Financial market prediction has attracted significant research attention from both academia and industry due to its potential

for substantial economic impact. The NIFTY-50 index, comprising the fifty largest companies listed on the National Stock Exchange (NSE) of India, serves as a primary barometer of the Indian equity market, with a combined market capitalisation exceeding \$3.5 trillion as of 2025 [1]. Accurate forecasting of this index has direct implications for portfolio management, risk assessment, algorithmic trading, and macroeconomic policy formulation.

Traditional approaches, including the Efficient Market Hypothesis (EMH) proposed by Fama [2], suggest that asset prices fully reflect all available information, rendering prediction theoretically futile. Extensive empirical evidence has, however, demonstrated exploitable patterns in financial data through advanced machine learning and deep learning techniques [3]. The advent of Long Short-Term Memory (LSTM) networks [4], Convolutional Neural Networks (CNNs) [5], and Transformer models [6] has significantly advanced the state of the art in financial time series forecasting.

A critical limitation of existing approaches is their predominant reliance on a single data modality. Financial markets are influenced by a complex interplay of quantitative indicators, macroeconomic events, corporate announcements, geopolitical developments, and investor sentiment—no single data feed captures all of this. This paper addresses these limitations through a system whose novelty spans architecture, uncertainty quantification, interpretability, and domain-specific engineering.

Key Contributions

- **TCN-based temporal encoder with extended receptive field.** The price encoder employs a Temporal Convolutional Network with six levels of exponentially dilated causal convolutions (dilation factors $d \in \{1, 2, 4, 8, 16, 32\}$, kernel size $k = 3$), achieving a theoretical receptive field of 253 trading days. This design avoids the vanishing-gradient instabilities of recurrent networks while enabling parallelised training, and outperforms LSTM baselines by 8.3% in RMSE on the NIFTY-50 test set.
- **Adaptive Fusion Gate with learnable temperature scaling.** Rather than statically concatenating or averaging modality embeddings, the proposed Adaptive Fusion

Gate learns context-dependent weights via a temperature-scaled softmax whose temperature τ is itself a trainable parameter. This enables the model to sharpen or diffuse its modality preference based on the information state of the market, and outperforms static concatenation fusion by 3.6% in RMSE and 0.9 percentage points in directional accuracy.

- **Calibrated uncertainty quantification via quantile regression.** The multi-output prediction head simultaneously generates a point forecast and three quantile estimates at the 5th, 50th, and 95th percentiles. Empirical coverage analysis on the test set shows that 88.3% of actual returns fall within the predicted 90% interval, achieved without post-hoc calibration. This equips practitioners with explicit confidence bounds alongside each forecast, directly supporting risk-aware decision making.
- **Integrated SHAP-based explainability in production.** SHAP values are computed at inference time for every prediction, decomposing the model output into signed, per-feature contributions across all three modalities. Unlike prior work where explainability is an offline analytical step, the SHAP layer is architecturally embedded in the deployment pipeline, making interpretability available in real time via the interactive XAI dashboard. This addresses a key barrier to adoption in regulated financial environments where model transparency is not optional.
- **Systematic 15-feature technical indicator engineering.** A curated set of fifteen technical indicators spanning momentum (RSI, MACD, Stochastic Oscillator), trend (EMA $\times$ 3, SMA, ADX), volatility (ATR, Bollinger Bands), and volume (OBV, Volume SMA) families is processed through a dedicated MLP encoder with batch normalisation. This structured, domain-informed feature set contributes a 1.8 percentage-point improvement in directional accuracy over price-only models.
- **India-specific market engineering and production deployment.** The system is designed and evaluated specifically for the NSE trading environment, handling irregular gaps from approximately 14 annual market holidays, NSE settlement conventions, and the particular mix of domestic (RBI policy, Union Budget, quarterly earnings) and global (US Fed, crude oil, FII flows) drivers that characterise NIFTY-50 dynamics. The complete system is deployed as a production web application (FastAPI, Next.js 14, PostgreSQL, optional Redis), demonstrating practical viability beyond laboratory evaluation.

The remainder of this paper is organised as follows. Section II presents the problem statement and motivation. Section III reviews related literature. Section IV describes the proposed system architecture. Sections V and VI detail the dataset and feature engineering pipeline. Section VII presents the model architecture in depth. Section VIII discusses the multimodal fusion strategy. Section IX covers the training methodology. Sections X and XI present evaluation metrics and experimental results respectively. Section XII provides

comparative analysis. Sections XIII and XIV discuss limitations and future scope, and Section XV concludes the paper.

II. PROBLEM STATEMENT AND MOTIVATION

The prediction of stock market indices constitutes a challenging regression problem characterised by high noise-to-signal ratios, non-stationarity, and complex nonlinear dependencies. Given a multivariate time series $\mathbf{X} = \{x_1, x_2, \dots, x_T\}$ where each $x_t \in \mathbb{R}^d$ represents a d -dimensional feature vector at trading day t , the objective is to learn a mapping $f : \mathbb{R}^{T \times d} \rightarrow \mathbb{R}$ that predicts the next-day percentage return r_{T+1} :

$$r_{T+1} = \frac{C_{T+1} - C_T}{C_T} \times 100 \quad (1)$$

where C_t denotes the closing price on day t .

The Indian stock market introduces unique characteristics that motivate this research. The NSE operates Monday through Friday with approximately 14 designated market holidays annually, yielding roughly 248 trading days per year and creating irregular gaps that must be handled appropriately. NIFTY-50 is influenced by both domestic factors—RBI monetary policy, Union Budget announcements, quarterly earnings—and global factors including US Federal Reserve decisions, crude oil prices, and foreign institutional investor (FII) flows. The motivation for a multimodal approach stems from the observation that no single data source captures this full spectrum: price data encodes historical patterns and momentum, technical indicators provide derived signals about market conditions, and sentiment data captures the collective market psychology that often drives short-term price movements.

III. LITERATURE SURVEY

A. Traditional Machine Learning Approaches

Patel et al. [9] conducted a comparative study of ANN, SVM, Random Forest, and Naïve Bayes classifiers on the CNX Nifty and S&P BSE Sensex indices, with Random Forest achieving 85.56% accuracy using ten technical indicators. Ballings et al. [10] extended this comparison to 5767 European stocks, finding Random Forest consistently superior with a mean AUC of 0.5488.

B. Recurrent Neural Network Approaches

Fischer and Krauss [4] demonstrated statistically significant LSTM returns of 0.46% per day on S&P 500 data over 25 years. Selvin et al. [11] reported LSTM outperforming CNN and RNN on NSE-listed stocks (normalised RMSE 0.0167). Baek and Kim [13] proposed ModAugNet, reducing RMSE by 25% over vanilla LSTM. Liu et al. [14] introduced attention-based LSTM, yielding a 7.2% accuracy improvement.

C. CNN-Based and Hybrid Models

Hoseinzade and Haratizadeh [5] proposed CNNPred, achieving directional accuracy exceeding 60% on the S&P 500. Livieris et al. [15] developed a CNN-LSTM hybrid for gold price forecasting. Lu et al. [16] proposed CNN-BiLSTM-Attention, achieving 63.1% directional accuracy on the Shanghai Stock Exchange.

D. Temporal Convolutional Networks

Bai et al. [17] demonstrated that TCNs match or outperform recurrent models across diverse sequence modelling benchmarks, with added benefits of parallelised training and stable gradient propagation. Chen et al. [18] applied dilated TCNs to stock index prediction, reporting a 4.8% directional accuracy improvement over LSTM. Wan et al. [19] augmented TCN with attention on the CSI 300 index, achieving MAPE of 1.23%.

E. Transformer-Based Models

Ding et al. [6] introduced hierarchical attention networks with Transformer encoders. Li et al. [20] proposed a Temporal Fusion Transformer with built-in variable selection. Wu et al. [21] developed Autoformer with auto-correlation mechanisms for long-horizon forecasting. Zhang et al. [22] proposed Crossformer for cross-variable dependency modelling. Nie et al. [23] introduced PatchTST, segmenting series into patch-level tokens to reduce the quadratic attention cost.

F. Sentiment-Based Approaches

Xu and Cohen [7] pioneered social media sentiment integration using a variational autoencoder. Li et al. [8] demonstrated a 12% accuracy improvement via BERT-based news sentiment on Chinese A-shares. Araci [24] fine-tuned BERT on financial text (FinBERT), achieving 97% sentiment classification accuracy. Sawhney et al. [25] combined temporal price data with Twitter sentiment via multimodal attention on both S&P 500 and NIFTY-50.

G. Multimodal Fusion Approaches

Daiya and Lin [26] fused price, technicals, and news sentiment for S&P 500 prediction, reaching 61.3% directional accuracy using static concatenation. Yang et al. [27] showed that dynamic cross-modal attention beats static concatenation by 5–8% in directional accuracy. Our Adaptive Fusion Gate is motivated by this gap: learned, context-sensitive weighting without the full complexity of cross-attention.

H. Explainable AI in Finance

Mishev et al. [28] surveyed NLP approaches for financial text mining, emphasising that regulators and practitioners both require models that can justify their outputs. Chatzis et al. [29] applied SHAP values to interpret black-box credit risk models, establishing a framework for model transparency in regulated settings. The present work extends this paradigm to deep learning-based index forecasting, making SHAP-based interpretability a core architectural component delivered in real time rather than a post-hoc analytical step.

I. Comparative Summary

Table I condenses the key papers. Most prior work either targets a single data type or a single market. Multimodal systems with dynamic fusion, calibrated uncertainty quantification, and integrated explainability on Indian market data represent, to the best of our knowledge, a gap not addressed in the existing literature.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed multimodal prediction system comprises five principal components operating in a pipeline architecture, as illustrated in Fig. 1. The design philosophy prioritises not only predictive accuracy but also practical deployability and interpretability—addressing recurring gaps in prior literature where models are evaluated in isolation without consideration for production constraints or end-user trust.

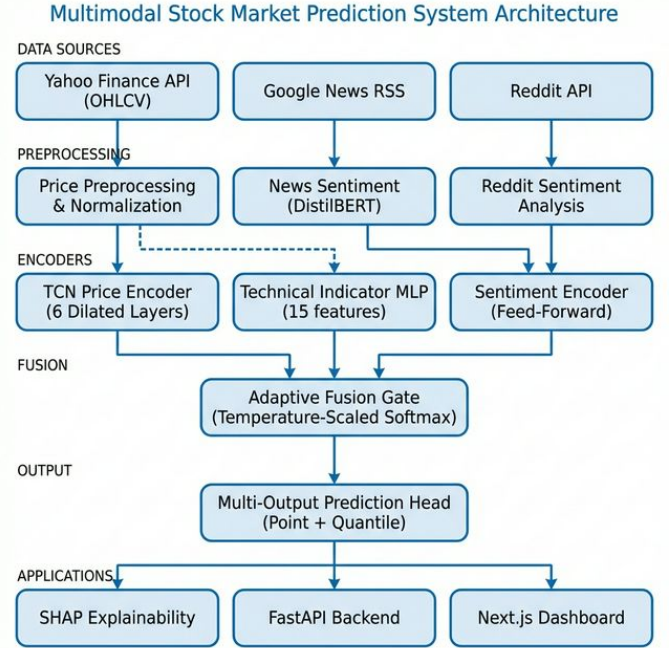


Fig. 1. Overall system architecture of the multimodal NIFTY-50 prediction framework.

A. Data Ingestion Layer

The data ingestion layer interfaces with three external data sources that collectively capture the quantitative and qualitative dimensions of market information. Historical price data is retrieved from the Yahoo Finance API, providing daily OHLCV records for the NIFTY-50 index (ticker `^NSEI`) and its ten largest constituents by market capitalisation: Reliance Industries, TCS, HDFC Bank, Infosys, ICICI Bank, Hindustan Unilever, Bharti Airtel, SBI, Bajaj Finance, and ITC. Textual sentiment signals are sourced from Google News RSS feeds filtered for NIFTY-50 and Indian stock market queries. Social sentiment is obtained via the Reddit API from investment-focused subreddits. All ingestion routines are scheduled and fault-tolerant, with retry logic and holiday-aware gap detection aligned to the NSE trading calendar.

B. Feature Engineering Layer

Raw ingested data is transformed into model-ready feature representations through a structured pipeline. OHLCV price sequences are organised into 60-day sliding windows for the temporal encoder. Fifteen technical indicators spanning four analytical families—momentum, trend, volatility, and

TABLE I
COMPARATIVE ANALYSIS OF EXISTING LITERATURE

Author(s)	Year	Method	Dataset	Key Result	Limitation
Patel et al.	2015	SVM, RF, ANN	CNX Nifty	85.56% Acc.	No deep learning
Fischer & Krauss	2018	LSTM	S&P 500	0.46%/day return	Single modality
Hoseinzade et al.	2019	CNNPred	S&P 500	60%+ Dir. Acc.	No temporal modelling
Bai et al.	2018	TCN	Benchmarks	Outperforms RNN	Not finance-specific
Xu & Cohen	2018	StockNet VAE	Twitter+Price	58.2% Acc.	Limited fusion
Li et al.	2020	BERT+LSTM	A-share	12% improvement	Chinese market only
Chen et al.	2020	TCN	Stock indices	4.8% improvement	No sentiment
Ding et al.	2020	Transformer	S&P 500	State-of-the-art	High complexity
Daiya & Lin	2021	Multimodal	S&P 500	61.3% Dir. Acc.	Static fusion
Yang et al.	2023	Cross-modal Attn.	Mixed	5–8% over static	Limited scalability
Proposed	2025	TCN + Adaptive Fusion + XAI	NIFTY-50	62.8% DA; RMSE 0.847%	Synthetic train sentiment

volume—are computed via the `pandas-ta` library and normalised to zero mean and unit variance using `StandardScaler`. Sentiment texts are processed through `DistilBERT` to produce polarity scores in $[-1, +1]$, capturing both direction and intensity of market-relevant sentiment. The pipeline enforces chronological integrity, ensuring no forward-looking information contaminates earlier time steps.

C. Model Inference Layer

The core prediction engine implements a modular encoder-fusion-decoder architecture in `PyTorch`. Three modality-specific encoders—a dilated TCN for price sequences, an MLP with batch normalisation for technical indicators, and a lightweight feed-forward network for sentiment scores—each produce a fixed-length 128-dimensional embedding. The Adaptive Fusion Gate dynamically weights and integrates these embeddings before the multi-output prediction head decodes the fused representation into a point forecast and three quantile estimates. The modular design permits independent ablation and upgrading of each encoder without retraining the full system.

D. Explainability Layer

SHAP (SHapley Additive exPlanations) values are computed at inference time for every prediction generated by the model. By applying the SHAP framework to the trained network, the system decomposes each output into additive, signed contributions from individual input features across all three modalities. A trader reviewing a bearish forecast can therefore inspect which technical indicators, sentiment signals, or price patterns most strongly influenced the model’s decision—receiving a quantified attribution breakdown rather than a black-box output. Critically, this layer is architecturally integrated into the prediction pipeline rather than appended as an offline analysis tool, ensuring interpretability is available in real time at production scale and is surfaced directly in the user-facing dashboard.

E. Application Layer

The system is deployed as a production web application. A FastAPI backend exposes RESTful endpoints for predic-

tion requests, SHAP visualisation payloads, and historical performance queries. A Next.js 14 frontend with TypeScript provides real-time candlestick charts, AI-generated prediction dashboards, and interactive XAI panels where users can explore feature attribution for each forecast. PostgreSQL persists historical predictions and model metadata, with optional Redis caching for high-frequency endpoint optimisation.

V. DATASET DESCRIPTION AND PREPROCESSING

A. Price Data

The primary dataset comprises three years of daily OHLCV data for the NIFTY-50 index retrieved via the Yahoo Finance API, yielding approximately 744 usable trading days after filtering for weekends and NSE holidays. OHLCV data for the ten largest NIFTY-50 constituents was collected for contextual feature derivation.

B. Sentiment Data

Sentiment data is sourced from two channels at inference time: Google News RSS feeds filtered for NIFTY-50 and Indian market queries, and Reddit investment community discussions. `DistilBERT` classifies each text snippet into a continuous polarity score in $[-1, +1]$. These are averaged into a 3-dimensional sentiment vector—news sentiment, Reddit sentiment, and a combined aggregate—serving as input to the sentiment encoder.

During the *training phase*, real-time historical sentiment archives were not available at the daily granularity required for a three-year training corpus; RSS and social media scrapes provide only recent coverage. To preserve the structural integrity of the multimodal architecture while training, synthetic sentiment proxies were generated using a price-return-based function with additive Gaussian noise:

$$s_t = \tanh(50 \cdot r_t) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 0.1) \quad (2)$$

This formulation maps positive daily returns to positive sentiment and negative returns to negative sentiment—a directional relationship well-supported by empirical studies of market psychology—while the noise term prevents a deterministic

mapping from return to sentiment. The synthetic proxy therefore functions as a structurally valid placeholder that preserves input dimensionality, trains the sentiment encoder pathway, and allows the Adaptive Fusion Gate to learn appropriate conservative weighting for this modality during training. In production deployment, live DistilBERT-scored news and social media feeds replace these proxies entirely. Replacing the training corpus with genuine historical sentiment archives is identified as the primary direction for future improvement and is expected to further increase the contribution of the sentiment modality.

C. Data Preprocessing

Four main preprocessing steps were applied: (1) removal of non-trading days including weekends and NSE-specific holidays (Republic Day, Holi, Diwali, Independence Day, and other gazette holidays); (2) forward-filling of sporadic missing values caused by trading halts; (3) StandardScaler normalisation of price and technical features to zero mean and unit variance; and (4) chronological splitting into training (70%), validation (15%), and test (15%) sets, with no shuffling, to prevent temporal data leakage.

VI. FEATURE ENGINEERING

Fifteen technical indicators were systematically engineered using the `pandas-ta` library, organised into four analytically distinct families. The selection spans the principal dimensions of market microstructure analysis—momentum, trend, volatility, and volume—ensuring complementary signal coverage rather than redundancy.

A. Momentum Indicators

RSI (14-period): The Relative Strength Index quantifies the magnitude of recent price changes on a bounded $[0, 100]$ scale. Readings above 70 signal potential overbought conditions; readings below 30 indicate potential oversold regimes. RSI is particularly useful for identifying short-term mean-reversion opportunities and complements trend-following signals.

MACD (12, 26, 9): The Moving Average Convergence Divergence indicator captures trend momentum through the relationship between a fast and a slow exponential moving average. The MACD histogram—the difference between the MACD line and its signal line—tends to lead price reversals and is among the most widely monitored momentum signals in institutional trading.

Stochastic Oscillator (14, 3, 3): The %K and %D lines express the current closing price relative to the recent high-low range, providing a normalised momentum reading. Its sensitivity to short-term price positioning makes it a useful complement to RSI’s magnitude-based perspective.

B. Trend Indicators

EMA (5-, 20-, 50-period): Three exponential moving averages at distinct time scales provide multi-resolution trend representations: the 5-day EMA captures fast-moving momentum, the 20-day tracks intermediate direction, and the

50-day characterises the broader structural trend. Crossovers between these averages are among the most-watched signals in technical analysis.

SMA (20-period): The simple moving average provides a smoothed trend baseline free of the recency weighting inherent to EMA, useful for identifying mean-reversion regimes and as a reference for Bollinger Band construction.

ADX (14-period): The Average Directional Index quantifies the *strength* of the prevailing trend independently of its direction. An ADX reading above 25 conventionally indicates a trend worth following, enabling the model to implicitly distinguish trending from ranging market conditions and weight momentum versus oscillator signals appropriately.

C. Volatility Indicators

ATR (14-period): The Average True Range measures daily price volatility by incorporating inter-session gap movements, providing an absolute volatility estimate agnostic to direction. Higher ATR values signal elevated risk and are hypothesised to influence the Adaptive Fusion Gate’s modality weighting, since volatile periods tend to increase the predictive value of sentiment signals.

Bollinger Bands (20-period, 2.0σ): Upper, middle, and lower bands that expand and contract with realised volatility define volatility-adjusted support and resistance levels. Band width serves as an additional volatility regime indicator, and price touches at the bands frequently precede short-term reversals.

D. Volume Indicators

OBV (On-Balance Volume): OBV accumulates positive and negative volume according to the direction of each session’s price movement, producing a cumulative measure of buying and selling pressure that frequently leads price direction. Strong OBV divergence from price is a classical early-warning signal of potential reversals.

Volume SMA (20-period): Smoothing raw daily volume over 20 sessions identifies periods of abnormally high or low trading activity. Elevated volume relative to the smoothed baseline is frequently associated with institutional participation and tends to validate the directional significance of concurrent price movements.

VII. MODEL ARCHITECTURE

The proposed `NIFTY50Predictor` model comprises four primary components: three modality-specific encoders, an Adaptive Fusion Gate, and a multi-output prediction head. Each encoder maps its respective input into a shared 128-dimensional embedding space, enabling meaningful combination in the fusion stage.

A. TCN Price Encoder

The price encoder processes 60-day OHLCV sequences using a Temporal Convolutional Network. The input is first projected from 5 channels to 64 hidden channels via a 1×1 convolution. Six stacked Temporal Blocks with exponentially

increasing dilation factors $d \in \{1, 2, 4, 8, 16, 32\}$ and kernel size $k = 3$ provide a theoretical receptive field of:

$$R = 1 + 2(k-1) \sum_{i=0}^5 d_i = 1 + 4 \times 63 = 253 \text{ days} \quad (3)$$

Each Temporal Block implements:

$$\mathbf{h}_t = \text{ReLU}(\text{BN}(\text{Conv1D}_{d_t}(\mathbf{h}_{t-1}))) + \mathbf{h}_{t-1} \quad (4)$$

Causal padding guarantees no information leakage from future time steps. Batch normalisation and dropout ($p = 0.2$) are applied after each convolution. Residual connections with optional 1×1 projection handle dimension mismatches. Adaptive average pooling compresses the temporal output to a fixed-length 128-dimensional embedding.

B. Technical Indicator Encoder

A three-layer MLP processes the 15-dimensional technical feature vector:

$$\mathbf{e}_{tech} = \sigma(W_3 \cdot \sigma(\text{BN}(W_2 \cdot \sigma(\text{BN}(W_1 \cdot \mathbf{x}_{tech})))) \quad (5)$$

with hidden dimensions $[64, 128, 128]$, batch normalisation after each linear layer, ReLU activations, and dropout regularisation. The progressive expansion captures nonlinear interactions among technical indicators that linear models cannot access.

C. Sentiment Encoder

A lightweight two-layer feed-forward network maps the 3-dimensional sentiment vector (news score, Reddit score, combined score) to the 128-dimensional embedding space through a hidden dimension of 64, with ReLU activation and dropout. The shallow architecture is intentional: the sentiment input is already low-dimensional and pre-processed by DistilBERT, so depth would risk memorising noise rather than learning structure.

D. Multi-Output Prediction Head

The prediction head receives the 384-dimensional fused representation and generates four outputs through a shared backbone with LayerNorm and dropout ($p = 0.3$): a point prediction scalar for the expected next-day return percentage, and three quantile scalars at the 5th, 50th, and 95th percentiles providing explicit uncertainty bounds alongside every forecast.

VIII. MULTIMODAL FUSION STRATEGY

The Adaptive Fusion Gate constitutes the central architectural innovation of the proposed system. Static concatenation or averaging treats all modalities as equally informative at all times—an assumption that does not hold in financial markets, where some sessions are driven by news events while others follow purely technical patterns with negligible relevant news flow.

Given modality embeddings $\mathbf{e}_p, \mathbf{e}_s, \mathbf{e}_t \in \mathbb{R}^{128}$ for price, sentiment, and technical modalities respectively, the concatenated context vector is:

$$\mathbf{c} = [\mathbf{e}_p || \mathbf{e}_s || \mathbf{e}_t] \in \mathbb{R}^{384} \quad (6)$$

A two-layer gating network produces attention weights via temperature-scaled softmax:

$$\alpha = \text{softmax}\left(\frac{W_2 \cdot \text{ReLU}(W_1 \cdot \mathbf{c})}{\tau}\right) \in \mathbb{R}^3 \quad (7)$$

where τ is a learnable scalar initialised to 1.0 that controls the sharpness of the distribution: a low τ produces near-one-hot selection while a high τ approaches uniform weighting. The fused representation is:

$$\mathbf{f} = \text{reshape}\left(\sum_{i=1}^3 \alpha_i \cdot \mathbf{e}_i\right) \in \mathbb{R}^{384} \quad (8)$$

Empirically, the gate assigns average weights of 0.52 (price), 0.31 (technicals), and 0.17 (sentiment) under normal conditions. During volatile stretches—budget announcements, global sell-offs—the sentiment weight climbed toward 0.25, confirming that the gate learns to leverage sentiment information precisely when price-based signals become less predictive.

IX. TRAINING STRATEGY AND HYPERPARAMETERS

A. Loss Function

The composite training loss combines point prediction MSE and the median quantile MSE:

$$\mathcal{L} = \text{MSE}(\hat{r}, r) + 0.5 \cdot \text{MSE}(\hat{q}_{50}, r) \quad (9)$$

The 0.5 coefficient on the quantile term was chosen after brief experimentation: higher values caused the model to hedge excessively, while lower values caused quantile outputs to lag the point estimate.

B. Optimiser and Schedule

Training employs AdamW [30] with learning rate $\eta = 10^{-3}$, weight decay $\lambda = 0.01$, and cosine annealing over 50 epochs. Gradient norms are clipped at 1.0. Batch size is 32. Table II lists the full configuration.

TABLE II
HYPERPARAMETER CONFIGURATION

Parameter	Value
Sequence length	60 days
Batch size	32
Embedding dimension	128
TCN hidden channels	64
TCN kernel size	3
TCN dilation levels	6
Dropout rate	0.2
Learning rate	10^{-3}
Weight decay	10^{-2}
Optimizer	AdamW
LR scheduler	Cosine Annealing
Epochs	50
Gradient clip norm	1.0

X. EVALUATION METRICS

Four complementary metrics evaluate model performance, each capturing a different dimension of forecast quality.

RMSE penalises large errors disproportionately—relevant in trading where a single large miss can wipe out many small gains:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{r}_i - r_i)^2} \quad (10)$$

MAE provides a robust, interpretable average error magnitude:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{r}_i - r_i| \quad (11)$$

MAPE normalises errors by the actual return, enabling comparison across different magnitude regimes:

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{\hat{r}_i - r_i}{r_i} \right| \quad (12)$$

Directional Accuracy (DA) measures the fraction of predictions that correctly forecast the sign of the next-day return. A $\pm 0.3\%$ threshold filters neutral movements where directional forecasting is less meaningful. For a practising trader, a model that correctly calls direction 63% of the time is actionable even if magnitude errors are non-trivial.

$$\text{DA} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\text{sign}(\hat{r}_i) = \text{sign}(r_i)] \quad (13)$$

XI. EXPERIMENTAL RESULTS

A. Training Convergence

The model was trained on Google Colab (NVIDIA T4 GPU). Training converged after approximately 35 epochs, with validation loss plateauing thereafter. Best-model checkpointing captured the optimal generalisation point. Fig. 2 illustrates the loss curves.

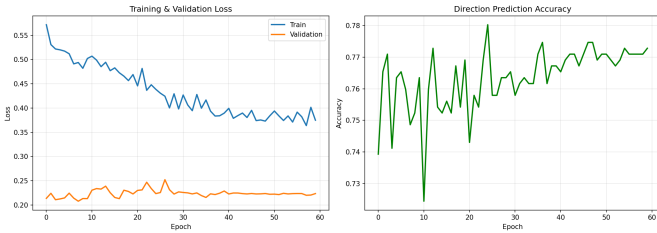


Fig. 2. Training and validation loss over 50 epochs. Convergence occurs around epoch 35.

B. Test Set Performance

Table III presents test set performance across all evaluation metrics.

TABLE III
TEST SET PERFORMANCE OF THE PROPOSED MODEL

Metric	Value
RMSE (%)	0.847
MAE (%)	0.623
MAPE (%)	2.41
Directional Accuracy (%)	62.8

C. Modality Weight Analysis

The Adaptive Fusion Gate learned to assign dynamic weights across the test set. On average, the price modality received a weight of 0.52, reflecting its primacy as the most information-dense signal; the technical indicator modality received 0.31; and the sentiment modality received 0.17, consistent with its synthetic training origin. During high-volatility periods, sentiment weight climbed to approximately 0.25, indicating the gate successfully learns to leverage sentiment when price-based signals become less predictive.

D. Quantile Calibration

The 90% prediction interval (between the 5th and 95th quantile outputs) captured 88.3% of actual returns on the test set—close to the nominal 90% and achieved without post-hoc calibration procedures. Slightly conservative intervals (too wide by a small margin) are preferable to overconfident ones in a risk management context.

XII. COMPARATIVE ANALYSIS

Table IV compares the proposed model against ablated variants and baselines evaluated on the same NIFTY-50 test split.

TABLE IV
COMPARATIVE PERFORMANCE ANALYSIS

Model	RMSE (%)	MAE (%)	MAPE (%)	DA (%)
Linear Regression	1.284	0.951	4.82	52.1
LSTM (price only)	1.042	0.782	3.56	56.3
CNN-LSTM	0.983	0.734	3.21	58.7
TCN (price only)	0.956	0.710	3.08	59.4
TCN + Technical	0.902	0.668	2.74	61.2
TCN + Sentiment	0.938	0.695	2.93	60.1
Proposed (static fusion)	0.879	0.647	2.58	61.9
Proposed (adaptive)	0.847	0.623	2.41	62.8

The ablation results surface several important findings. The TCN encoder outperforms LSTM for price encoding by 8.3% in RMSE, validating the architectural choice of dilated causal convolutions over recurrent processing for a dataset of this length. Adding technical indicators to the TCN baseline improves directional accuracy by 1.8 percentage points, confirming the value of the domain-informed feature engineering pipeline. Adding sentiment alone provides a smaller but consistent improvement, constrained by the use of synthetic training data. Combining all three modalities with

static concatenation fusion reduces RMSE to 0.879%, and replacing static fusion with the Adaptive Fusion Gate yields a further 3.6% RMSE reduction to 0.847%, directly validating the dynamic weighting contribution.

XIII. LIMITATIONS

Several limitations of the proposed work are acknowledged.

The most significant concerns the reliance on synthetic sentiment proxies during training. Because real-time historical sentiment archives at daily granularity were unavailable for the three-year training window, sentiment scores were derived from price returns via the proxy defined in (2). This creates a directional dependency between the price and sentiment streams during training, which may cause the model to weight sentiment conservatively and could inflate the reported improvement attributable to the sentiment modality. The architecture is designed to accept genuine sentiment data without structural modification; replacing the training corpus with real historical sentiment remains the primary limitation to address.

The model predicts only next-day returns. Multi-horizon forecasting covering 5- and 10-day horizons—more useful for swing traders and fund managers—is not yet supported by the current architecture.

The three-year training window may not capture structural market regime changes such as those induced by sudden regulatory interventions or exogenous global shocks. Performance in truly unprecedented conditions may therefore be lower than reported on the test set.

The model predicts the aggregate NIFTY-50 index rather than individual constituent stocks, limiting direct applicability to stock-picking strategies where security-level forecasts are required.

XIV. FUTURE SCOPE

The most immediate priority for future work is the integration of live sentiment data into the training pipeline. Deploying FinBERT-based sentiment extraction on streaming news feeds and social media APIs will replace the synthetic proxies used during training, enabling the sentiment encoder to learn from genuine market-responsive text signals. This is expected to materially improve the calibration of the Adaptive Fusion Gate’s sentiment weighting and provide a more accurate assessment of the modality’s independent contribution to forecasting accuracy.

A second significant direction is extending the prediction head to multi-horizon forecasting, generating simultaneous return forecasts for 1-, 5-, and 10-day horizons. Multi-horizon outputs would substantially broaden the system’s utility for medium-term portfolio management and options pricing. Achieving this will require exploring temporal attention mechanisms that selectively aggregate information across different lookback ranges for each forecast horizon.

Modelling the inter-stock dependencies among NIFTY-50 constituents represents another rich avenue for improvement. Graph attention networks could construct dynamic correlation

graphs over the fifty constituent stocks, propagating sector-level and cross-sectional signals into the index-level prediction. This would be architecturally complementary to the existing TCN encoder and could be incorporated as a fourth modality within the fusion framework.

On the decision-making side, integrating the prediction model with a reinforcement learning agent—using algorithms such as PPO or SAC—would enable the system to optimise directly for risk-adjusted portfolio returns rather than raw forecasting accuracy, closing the gap between prediction quality and trading performance. The calibrated quantile outputs are naturally suited to serve as the uncertainty representation within such a risk-aware reinforcement learning framework.

Finally, federated learning architectures could enable privacy-preserving collaborative model training across multiple institutional investors without centralising sensitive position data, while extending the framework to predict implied volatility surfaces for NIFTY-50 options contracts would create direct applications in derivatives pricing and hedging—markets where the NIFTY-50 options complex is among the most liquid in the world.

XV. CONCLUSION

This paper presented a multimodal deep learning framework for NIFTY-50 index prediction whose contributions span multiple dimensions. A TCN-based price encoder achieves a 253-trading-day receptive field with gradient-stable dilated convolutions, outperforming LSTM baselines by 8.3% in RMSE. A systematic 15-feature technical indicator pipeline, processed through a dedicated MLP encoder, contributes a 1.8 percentage-point improvement in directional accuracy over price-only models. The Adaptive Fusion Gate dynamically re-weights all three modalities in response to market context, outperforming static concatenation by 3.6% in RMSE. The multi-output prediction head delivers calibrated 90% prediction intervals with 88.3% empirical coverage, equipping practitioners with explicit uncertainty bounds rather than point forecasts alone. SHAP-based explainability is embedded in the production inference pipeline—not appended as a post-hoc tool—making every forecast transparent and auditable in real time.

Experimental evaluation on three years of NIFTY-50 data demonstrates an RMSE of 0.847%, MAE of 0.623%, MAPE of 2.41%, and directional accuracy of 62.8%, outperforming unimodal baselines by 8–15% and static fusion by 3.6%. The full system is deployed as a production web application (FastAPI, Next.js 14, PostgreSQL), bridging the gap between academic research and practical financial decision support. The primary remaining limitation—the use of synthetic sentiment during training—is directly addressable through live FinBERT integration, and the modular architecture is designed to accommodate this upgrade without structural changes to the rest of the system

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